ICMP Covert Channel Detection - Feature Summary Table

Quick Reference: Required Flow-Level Features by Anti-Forensic Technique

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1. TIME OBFUSCATION FEATURES (29 total)

#	Feature Name	Priority	Description	Derivable from Current Data
1	flow_duration	HIGH	Total flow duration	✅ Yes
2	flow_packets_per_second	HIGH	Packet rate	✅ Yes
3	flow_bytes_per_second	HIGH	Byte throughput	✅ Yes
4	avg_inter_arrival_time	HIGH	Mean packet timing	✅ Yes
5	std_inter_arrival_time	HIGH	Timing variation	✅ Yes
6	min_inter_arrival_time	MEDIUM	Minimum gap	✅ Yes
7	max_inter_arrival_time	MEDIUM	Maximum gap	✅ Yes
8	median_inter_arrival_time	MEDIUM	Median timing	✅ Yes
9	inter_arrival_time_variance	HIGH	Timing jitter	✅ Yes
10	inter_arrival_time_cv	MEDIUM	Coefficient of variation	✅ Yes
11	timing_entropy	HIGH	Timing randomness	✅ Yes
12	timing_skewness	MEDIUM	Distribution skew	✅ Yes
13	timing_kurtosis	MEDIUM	Outlier detection	✅ Yes
14	burst_count	HIGH	Number of bursts	✅ Yes
15	burst_ratio	HIGH	Burst percentage	✅ Yes
16	max_burst_size	MEDIUM	Max consecutive burst	✅ Yes
17	avg_burst_size	MEDIUM	Average burst size	✅ Yes
18	burst_duration	LOW	Time in bursts	✅ Yes
19	delay_spike_count	MEDIUM	Large delay count	✅ Yes
20	delay_spike_ratio	MEDIUM	Delay percentage	✅ Yes

#	Feature Name	Priority	Description	Derivable from Current Data
21	max_delay	LOW	Maximum delay	✅ Yes
22	delay_variance	LOW	Delay variation	✅ Yes
23	timing_autocorrelation	LOW	Periodic patterns	✅ Yes
24	timing_periodicity_score	LOW	Regularity measure	✅ Yes
25	timing_regularity_index	LOW	Timing consistency	✅ Yes
26	timing_variance_first_10	LOW	Early variance	✅ Yes
27	timing_variance_last_10	LOW	Late variance	✅ Yes
28	timing_variance_ratio	LOW	Variance change	✅ Yes
29	rolling_avg_timing_std	LOW	Rolling std dev	✅ Yes

Summary: 29 features, ALL derivable from current data ✅

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2. HEADER MANIPULATION FEATURES (40 total)

#	Feature Name	Priority	Description	Derivable from Current Data
30	icmp_id_count	HIGH	Unique IDs	✅ Yes (from Info)
31	icmp_id_entropy	HIGH	ID randomness	✅ Yes (from Info)
32	icmp_id_variance	MEDIUM	ID variation	✅ Yes (from Info)
33	icmp_id_pattern_score	MEDIUM	ASCII encoding	✅ Yes (from Info)
34	icmp_id_mode	LOW	Most common ID	✅ Yes (from Info)
35	icmp_id_change_count	HIGH	ID changes	✅ Yes (from Info)
36	icmp_id_change_rate	MEDIUM	Change frequency	✅ Yes (from Info)
37	icmp_seq_count	MEDIUM	Unique sequences	✅ Yes (from Info)
38	icmp_seq_min	LOW	Min sequence	✅ Yes (from Info)
39	icmp_seq_max	LOW	Max sequence	✅ Yes (from Info)
40	icmp_seq_range	MEDIUM	Sequence range	✅ Yes (from Info)
41	icmp_seq_mean	LOW	Mean sequence	✅ Yes (from Info)

#	Feature Name	Priority	Description	Derivable from Current Data
42	icmp_seq_std	MEDIUM	Sequence std dev	✅ Yes (from Info)
43	icmp_seq_sequential_ratio	HIGH	Normal behavior	✅ Yes (from Info)
44	icmp_seq_non_sequential_count	HIGH	Anomaly count	✅ Yes (from Info)
45	icmp_seq_reverse_count	MEDIUM	Reverse sequences	✅ Yes (from Info)
46	icmp_seq_duplicate_count	MEDIUM	Duplicate seq	✅ Yes (from Info)
47	icmp_seq_gap_count	MEDIUM	Missing sequences	✅ Yes (from Info)
48	icmp_seq_entropy	HIGH	Seq randomness	✅ Yes (from Info)
49	icmp_seq_randomness_score	MEDIUM	Randomness measure	✅ Yes (from Info)
50	icmp_type_count	HIGH	Type diversity	⚠️ Partial (from Info)
51	icmp_code_count	MEDIUM	Code diversity	⚠️ Partial (from Info)
52	icmp_type_mode	LOW	Common type	⚠️ Partial (from Info)
53	icmp_code_mode	LOW	Common code	⚠️ Partial (from Info)
54	icmp_type_entropy	MEDIUM	Type randomness	⚠️ Partial (from Info)
55	icmp_echo_request_ratio	MEDIUM	Request percentage	⚠️ Partial (from Info)
56	icmp_echo_reply_ratio	LOW	Reply percentage	⚠️ Partial (from Info)
57	icmp_request_reply_ratio	LOW	Request/reply ratio	⚠️ Partial (from Info)
58	ttl_count	MEDIUM	Unique TTLs	✅ Yes (from Info)
59	ttl_mean	LOW	Mean TTL	✅ Yes (from Info)
60	ttl_std	MEDIUM	TTL variation	✅ Yes (from Info)
61	ttl_min	LOW	Min TTL	✅ Yes (from Info)
62	ttl_max	LOW	Max TTL	✅ Yes (from Info)
63	ttl_range	LOW	TTL range	✅ Yes (from Info)
64	ttl_variance_score	MEDIUM	TTL consistency	✅ Yes (from Info)
65	ttl_mode	LOW	Common TTL	✅ Yes (from Info)
66	ttl_anomaly_count	LOW	Unusual TTLs	✅ Yes (from Info)

#	Feature Name	Priority	Description	Derivable from Current Data
67	icmp_checksum_variance	LOW	Checksum variation	❌ No - need full capture
68	icmp_checksum_entropy	LOW	Checksum randomness	❌ No - need full capture
69	icmp_checksum_anomaly_count	LOW	Invalid checksums	❌ No - need full capture

Summary: 40 features, 37 derivable (92%) ✅, 3 need full packet capture ❌

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3. PAYLOAD SIZE COVERT CHANNEL FEATURES (35 total)

#	Feature Name	Priority	Description	Derivable from Current Data
70	total_bytes	HIGH	Total flow bytes	✅ Yes
71	total_packets	HIGH	Total packet count	✅ Yes
72	avg_packet_size	HIGH	Mean size	✅ Yes
73	std_packet_size	HIGH	Size std dev	✅ Yes
74	min_packet_size	MEDIUM	Minimum size	✅ Yes
75	max_packet_size	MEDIUM	Maximum size	✅ Yes
76	median_packet_size	MEDIUM	Median size	✅ Yes
77	packet_size_range	MEDIUM	Size range	✅ Yes
78	packet_size_variance	HIGH	Size variation	✅ Yes
79	packet_size_cv	MEDIUM	Coefficient of variation	✅ Yes
80	packet_size_entropy	HIGH	Size randomness	✅ Yes
81	packet_size_skewness	MEDIUM	Distribution skew	✅ Yes
82	packet_size_kurtosis	MEDIUM	Outlier measure	✅ Yes
83	unique_packet_sizes	HIGH	Size diversity	✅ Yes
84	size_diversity_ratio	MEDIUM	Uniqueness ratio	✅ Yes
85	standard_size_ratio	HIGH	Normal size %	✅ Yes
86	non_standard_size_count	HIGH	Anomaly count	✅ Yes
87	non_standard_size_ratio	HIGH	Anomaly %	✅ Yes

#	Feature Name	Priority	Description	Derivable from Current Data
88	oversized_packet_count	MEDIUM	Large packet count	✅ Yes
89	oversized_packet_ratio	MEDIUM	Large packet %	✅ Yes
90	undersized_packet_count	MEDIUM	Small packet count	✅ Yes
91	undersized_packet_ratio	LOW	Small packet %	✅ Yes
92	size_mode	LOW	Most common size	✅ Yes
93	size_mode_frequency	LOW	Mode frequency	✅ Yes
94	size_alternation_count	MEDIUM	Size changes	✅ Yes
95	size_alternation_rate	MEDIUM	Change frequency	✅ Yes
96	consecutive_same_size_max	LOW	Max same-size run	✅ Yes
97	size_change_variance	LOW	Change variation	✅ Yes
98	avg_payload_size	MEDIUM	Mean payload	⚠️ Estimate (Length-28)
99	payload_size_variance	MEDIUM	Payload variation	⚠️ Estimate (Length-28)
100	payload_size_entropy	MEDIUM	Payload randomness	⚠️ Estimate (Length-28)
101	payload_to_header_ratio	LOW	Payload/header	⚠️ Estimate (Length-28)
102	variable_payload_ratio	LOW	Varying payload %	⚠️ Estimate (Length-28)
103	size_time_correlation	MEDIUM	Size-timing correlation	✅ Yes
104	large_packet_burst_ratio	LOW	Large in bursts	✅ Yes
105	small_packet_burst_ratio	LOW	Small in bursts	✅ Yes

Summary: 35 features, 30 fully derivable (86%) ✅, 5 estimable ⚠️

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4. FLOW BENDING FEATURES (69 total)

#	Feature Name	Priority	Description	Derivable from Current Data
106	ip_version	LOW	IP version	❌ No
107	ip_header_length_min	MEDIUM	Min header length	❌ No
108	ip_header_length_max	MEDIUM	Max header length	❌ No

#	Feature Name	Priority	Description	Derivable from Current Data
109	ip_header_length_avg	MEDIUM	Avg header length	❌ No
110	ip_header_length_variance	HIGH	Header variation	❌ No
111	ip_header_length_unique	LOW	Unique lengths	❌ No
112	ip_df_flag_count	HIGH	Don't Fragment count	❌ No
113	ip_mf_flag_count	HIGH	More Fragments count	❌ No
114	ip_df_ratio	HIGH	DF flag %	❌ No
115	ip_mf_ratio	HIGH	MF flag %	❌ No
116	ip_flag_variance	MEDIUM	Flag variation	❌ No
117	fragmented_packet_count	HIGH	Fragment count	❌ No
118	fragmentation_ratio	HIGH	Fragmentation %	❌ No
119	fragment_offset_max	MEDIUM	Max offset	❌ No
120	fragment_offset_variance	MEDIUM	Offset variation	❌ No
121	unique_fragment_ids	LOW	Unique frag IDs	❌ No
122	avg_fragments_per_id	LOW	Avg frags/ID	❌ No
123	incomplete_fragment_count	MEDIUM	Incomplete frags	❌ No
124	overlapping_fragment_count	MEDIUM	Overlapping frags	❌ No
125	ip_options_present_count	HIGH	Options count	❌ No
126	ip_options_ratio	HIGH	Options %	❌ No
127	ip_options_length_avg	MEDIUM	Avg option length	❌ No
128	ip_options_type_count	LOW	Option types	❌ No
129	ip_options_entropy	MEDIUM	Option randomness	❌ No
130	ip_timestamp_option_count	LOW	Timestamp options	❌ No
131	ip_record_route_count	LOW	Record route	❌ No
132	ip_source_route_count	MEDIUM	Source routing	❌ No
133	ip_id_min	LOW	Min IP ID	❌ No

#	Feature Name	Priority	Description	Derivable from Current Data
134	ip_id_max	LOW	Max IP ID	❌ No
135	ip_id_range	MEDIUM	IP ID range	❌ No
136	ip_id_sequential_ratio	MEDIUM	Sequential IDs	❌ No
137	ip_id_entropy	MEDIUM	ID randomness	❌ No
138	ip_id_randomness_score	LOW	Randomness score	❌ No
139	tos_count	MEDIUM	ToS diversity	❌ No
140	tos_mode	LOW	Common ToS	❌ No
141	tos_variance	LOW	ToS variation	❌ No
142	tos_anomaly_count	LOW	Unusual ToS	❌ No
143	ecn_flag_count	LOW	ECN flags	❌ No
144	unique_src_ips	MEDIUM	Source IP count	✅ Yes
145	unique_dst_ips	MEDIUM	Dest IP count	✅ Yes
146	src_ip_entropy	MEDIUM	Source randomness	✅ Yes
147	dst_ip_entropy	MEDIUM	Dest randomness	✅ Yes
148	src_ip_change_count	MEDIUM	Source changes	✅ Yes
149	dst_ip_change_count	LOW	Dest changes	✅ Yes
150	bidirectional_flag	LOW	Both directions	✅ Yes
151	forward_packets	MEDIUM	Forward count	✅ Yes
152	backward_packets	MEDIUM	Backward count	✅ Yes
153	forward_backward_ratio	MEDIUM	Direction ratio	✅ Yes
154	src_port_entropy	LOW	Source port (N/A ICMP)	❌ N/A
155	dst_port_entropy	LOW	Dest port (N/A ICMP)	❌ N/A
156	src_subnet_count	LOW	Source subnets	✅ Yes
157	dst_subnet_count	LOW	Dest subnets	✅ Yes
158	cross_subnet_flag	LOW	Cross subnet	✅ Yes

#	Feature Name	Priority	Description	Derivable from Current Data
159	flow_pkts_per_sec	HIGH	Packet rate	✅ Yes
160	flow_bytes_per_sec	HIGH	Byte rate	✅ Yes
161	flow_active_time	MEDIUM	Active duration	✅ Yes
162	flow_idle_time	MEDIUM	Idle duration	✅ Yes
163	flow_active_ratio	LOW	Active %	✅ Yes
164	flow_idle_count	LOW	Idle periods	✅ Yes
165	flow_symmetry_score	MEDIUM	Traffic symmetry	✅ Yes
166	flow_consistency_score	LOW	Pattern consistency	✅ Yes
167	flow_anomaly_score	MEDIUM	Overall anomaly	✅ Yes
168	ttl_hop_count_estimate	MEDIUM	Hop estimate	✅ Yes (from TTL)
169	ttl_consistency_score	MEDIUM	TTL consistency	✅ Yes (from TTL)
170	path_change_indicator	LOW	Path changes	⚠️ Partial (from TTL)
171	routing_loop_indicator	LOW	Loop detection	⚠️ Partial (from TTL)
172	protocol_type	LOW	Protocol (ICMP=1)	✅ Yes
173	protocol_variance	LOW	Protocol changes	✅ Yes

Summary: 69 features, 29 derivable (42%) ✅, 38 require full packet capture ❌, 2 N/A for ICMP

OVERALL SUMMARY

Technique	Total Features	Derivable from Current Data	Percentage
Time Obfuscation	29	29 ✅	100%
Header Manipulation	40	37 ✅ + 3 ❌	92%
Payload Size Covert	35	30 ✅ + 5 ⚠️	86%
Flow Bending	69	29 ✅ + 38 ❌ + 2 N/A	42%
TOTAL	**173**	**125 ✅ (72%)**	**72%**

Legend:

- ✅ **Fully derivable** from your current packet-level data
- ⚠️ **Estimable** (can approximate but not exact)
- ❌ **Requires full packet capture** (need IP/ICMP headers)
- N/A **Not applicable** for ICMP protocol

REALISTIC FEATURE SET FOR YOUR CURRENT DATASET

Immediately Available: ~95 features (High confidence)

- All 29 Time Obfuscation features
- 37 Header Manipulation features (from Info field parsing)
- 30 Payload Size features (from Length field)
- 29 Flow Bending features (flow statistics only)

With Estimates: ~100 features (Medium confidence)

- Add 5 payload estimation features

Requires Re-capture: ~73 features

- 38 IP-layer Flow Bending features
- 3 ICMP checksum features

RECOMMENDATION FOR YOUR FYP

OPTION 1: Work with Current Data (Recommended)

- Extract **95-100 flow-level features**
- Focus on Time Obfuscation + Header Manipulation + Payload Covert
- Acknowledge Flow Bending limitations in methodology
- **Advantage:** Can start immediately
- **Disadvantage:** Limited flow bending detection

OPTION 2: Re-capture with Full Headers

- Use `tshark` or `tcpdump` with full packet capture
- Extract all **173 features**
- Complete coverage of all 4 techniques

- **Advantage:** Comprehensive dataset
- **Disadvantage:** Need to regenerate steganographic traffic

OPTION 3: Hybrid Approach (Best)

- Phase 1: Use current data for 95 features
- Phase 2: Re-capture small validation set with full headers
- Compare results and validate findings
- **Advantage:** Balanced approach
- **Disadvantage:** More work

PRIORITY FEATURE LIST (Top 30 for Quick Start)

Rank	Feature	Technique	Priority
1	avg_inter_arrival_time	Time Obfuscation	HIGH
2	std_inter_arrival_time	Time Obfuscation	HIGH
3	burst_ratio	Time Obfuscation	HIGH
4	timing_entropy	Time Obfuscation	HIGH
5	avg_packet_size	Payload Covert	HIGH
6	std_packet_size	Payload Covert	HIGH
7	packet_size_entropy	Payload Covert	HIGH
8	non_standard_size_ratio	Payload Covert	HIGH
9	icmp_id_entropy	Header Manipulation	HIGH
10	icmp_id_change_count	Header Manipulation	HIGH
11	icmp_seq_sequential_ratio	Header Manipulation	HIGH
12	icmp_seq_entropy	Header Manipulation	HIGH
13	icmp_seq_non_sequential_count	Header Manipulation	HIGH
14	flow_duration	Time Obfuscation	HIGH
15	flow_packets_per_second	Flow Stats	HIGH
16	total_packets	Flow Stats	HIGH

Rank	Feature	Technique	Priority
17	total_bytes	Flow Stats	HIGH
18	unique_packet_sizes	Payload Covert	HIGH
19	inter_arrival_time_variance	Time Obfuscation	HIGH
20	icmp_type_count	Header Manipulation	HIGH
21	max_inter_arrival_time	Time Obfuscation	MEDIUM
22	packet_size_variance	Payload Covert	HIGH
23	size_time_correlation	Combined	MEDIUM
24	timing_skewness	Time Obfuscation	MEDIUM
25	icmp_id_count	Header Manipulation	HIGH
26	unique_src_ips	Flow Bending	MEDIUM
27	unique_dst_ips	Flow Bending	MEDIUM
28	ttl_variance_score	Header Manipulation	MEDIUM
29	delay_spike_count	Time Obfuscation	MEDIUM
30	oversized_packet_ratio	Payload Covert	MEDIUM

These 30 features alone should give you strong classification performance for ICMP covert channel detection!